PROJECT CHARTER

Customer Intelligence & Predictive Retention Analytics Engine

CIRPA-2026-S1

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Document Control

Version History

Version	Date	Author	Change Description
1.0	Feb 2026	Mai	Initial charter draft
2.0	Feb 2026	Mai	Restructured to Big 4 standard; added governance, WBS, RACI, quality gates, risk matrix
3.0	Feb 2026	Mai	Final professional edition; added communication plan, benefits realisation, stakeholder analysis, Gantt timeline, expanded methodology

Distribution

Role	Name	Access Level
Project Owner	Mai	Full
Technical Advisor	Claude (AI)	Consultation
Domain Reviewer	TBD (peer / faculty)	Review

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1. Executive Summary

This charter authorises the **Customer Intelligence & Predictive Retention Analytics Engine (CIRPA)** — a repeatable customer analytics workflow that segments an e-commerce retailer's customer base, predicts individual churn probability and lifetime value, diagnoses the drivers of attrition, and produces prioritised retention recommendations with quantified financial impact.

The system is designed as a recurring operational pipeline — not a one-time analysis — mirroring the customer intelligence cycles operated by Starbucks (loyalty personalisation), Kroger (precision marketing), Amazon (behavioural recommendation), and Netflix (engagement-driven retention).

Dimension	Detail
Dataset	UCI Online Retail II — 1,067,371 real transactions from a UK retailer (Dec 2009–Dec 2011)
ML Coverage	Unsupervised clustering + Supervised classification + Supervised regression + Tree-based ensembles
Business Output	Executive BI dashboard with segment profiles, churn risk scores, CLV predictions, retention ROI
Timeline	Semester 1, 2026 — 5 phases, ~127 hours total effort
Primary Audience	Big 4 consulting interviewers; data science hiring managers; portfolio reviewers
Strategic Extension	Semester 2: prescriptive optimisation layer (post-COMPSCI 225)

In one sentence:

Segment customers by behaviour, predict who will leave and what they are worth, explain why, and recommend what to do about it — on a repeatable schedule.

2. Business Case & Strategic Alignment

2.1 Industry Context

Customer retention economics are well-established across enterprise analytics. The following data points frame the commercial imperative:

Metric	Value	Source
Customer acquisition cost increase (5-year trend)	+60%	Harvard Business Review
Retention cost advantage vs. acquisition	5–25× cheaper	Harvard Business Review
Profit uplift from 5% retention increase	25–95%	Bain & Company
Global annual revenue lost to churn	>US$1.6 trillion	Accenture
Customer analytics share of retail big data revenue (2024)	34% (largest segment)	Mordor Intelligence
Predictive analytics share of retail analytics revenue (2024)	40.2%	Mordor Intelligence
Amazon recommendation engine revenue contribution	35% of total revenue	McKinsey Digital

2.2 Strategic Alignment

This project exists at the intersection of three strategic forces:

Force	Alignment
Enterprise Demand	78% of organisations deploy AI/ML in at least one function (McKinsey, 2025). Customer intelligence and prediction are the two most common deployments. This project builds the exact capability enterprises are investing in.
Career Positioning	Big 4 firms (target: Deloitte Engineering, AI & Data) deliver customer analytics as a core consulting service. This project produces a deliverable indistinguishable from a real engagement output.
Skill Demonstration	Among 10 use-case pairings evaluated, Prediction + Customer Intelligence is the only combination that naturally exercises all three intermediate ML categories — unsupervised, supervised, and tree-based — in an integrated workflow.

2.3 Benefits Realisation Plan

Benefits are categorised as tangible (directly measurable) or intangible (indirectly valuable), each with a defined realisation timeline.

ID	Benefit	Type	Measure	Realisation
B1	Portfolio piece for Big 4 internship applications	Tangible	Included in ≥1 application	Semester 1 end
B2	Interview-ready project walkthrough	Tangible	Can deliver 5-min + 15-min versions	Semester 1 end

B3	Validated competency across full DS lifecycle	Tangible	13 deliverables completed to quality gate standard	Semester 1 end
B4	Foundation for Semester 2 prescriptive extension	Tangible	Prediction models available for optimisation input	Semester 2 start
B5	Differentiation from standard Kaggle-style notebooks	Intangible	Peer review confirms consulting-grade framing	Semester 1 end
B6	Deepened understanding of enterprise analytics lifecycle	Intangible	Ability to discuss end-to-end pipeline in interviews	Ongoing
B7	Business communication skill development	Intangible	ROI quantification + executive dashboard delivery	Phase 4–5

3. Problem Statement

An e-commerce retailer possesses two years of granular transactional data but lacks the analytical infrastructure to convert that data into actionable customer intelligence. Four critical business questions remain unanswered:

#	Question	Business Consequence of Not Answering
Q1	Who are our customers? No behavioural segmentation exists.	All customers treated identically; marketing spend is unoptimised
Q2	Who is about to leave? No early-warning system for at-risk customers.	Churn discovered only after it occurs — when intervention is too late
Q3	What are they worth? No forward-looking CLV estimate.	Resource allocation ignores customer value; high-value customers under-served
Q4	What should we do about it? No framework to prioritise retention actions.	Blanket campaigns waste budget on low-value or already-loyal customers

This project delivers the analytical capability to answer all four questions within a single, integrated, repeatable workflow.

4. Project Objectives

All objectives follow SMART criteria: Specific, Measurable, Achievable, Relevant, Time-bound.

ID	Objective	Measure	Target	Phase
O1	Segment the customer base into distinct behavioural groups	Segments with statistically significant inter-group differences ($p < 0.05$ on ≥3 RFM dimensions)	3–5 segments	2
O2	Predict individual customer churn on unseen future data	AUC-ROC on temporal holdout	≥ 0.80	3
O3	Predict individual customer lifetime value	R^2 on temporal holdout (or accuracy ≥ 0.75 if pivoted to classification)	≥ 0.50	3
O4	Identify the top churn drivers per segment	Drivers explainable in plain English via SHAP	≥ 5 per segment	4
O5	Quantify financial impact of targeted retention	ROI expressed as £ preserved per £ spent	> 1:1	4
O6	Deliver an executive dashboard for non-technical stakeholders	5-second rule: primary insight visible immediately	Pass/Fail	5
O7	Document the workflow as a repeatable pipeline	Architecture diagram: input → process → output → feedback	Complete	5

5. Scope Definition

5.1 In-Scope Deliverables

ID	Deliverable	Phase	Analytics Purpose	ML Category
D1	Data ingestion & quality pipeline	1	Foundation	Data engineering
D2	Exploratory data analysis report	1	Descriptive (Visibility)	Statistical profiling
D3	RFM + extended feature engineering	1	Feature creation	Feature engineering
D4	Customer segmentation models	2	Customer Intelligence	Unsupervised
D5	Segment profiling with statistical validation	2	Diagnosis	Hypothesis testing
D6	Churn label definition & engineering	2	Prediction setup	Domain logic
D7	Churn prediction model suite	3	Prediction	Supervised + Tree-based
D8	CLV prediction model	3	Prediction	Supervised regression
D9	SHAP feature importance & driver analysis	4	Diagnosis	Model interpretability
D10	Retention priority matrix	4	Prescription-lite	Business scoring
D11	Financial impact & ROI quantification	4	Business case	ROI estimation
D12	Executive BI dashboard	5	Visibility (output)	Visualisation
D13	Workflow documentation & portfolio packaging	5	Operationalisation	Pipeline design

5.2 Out of Scope

Item	Rationale	Future Consideration
Real-time model serving / API deployment	Infrastructure beyond portfolio requirements	Semester 2 — MLOps path
Deep learning / neural networks	Outside curriculum; unnecessary for tabular retail data	Not planned — tree-based models are industry standard here
Mathematical optimisation (LP, simulation)	Requires COMPSCI 225 (Semester 2)	Semester 2 — prescriptive analytics extension
A/B test design for campaigns	Requires live production environment	Documented as recommended next step
External data enrichment	Single-dataset constraint	Noted as enterprise extension in limitations

5.3 Scope Boundaries

Dimension	Boundary	Rationale
Temporal	Training: Dec 2009 – Jun 2011 | Holdout: Jul – Dec 2011	Simulates production forecasting on unseen future data
Geographical	UK customers only (91% of transactions, ~85% of revenue)	Statistical validity; international noted as extension
Customer filter	Valid CustomerID only (~75% of records)	Null IDs unattributable; bias tested in EDA
Product filter	Exclude non-product codes (POST, DOT, M, BANK CHARGES, CRUK)	Operational entries, not customer purchases

6. Dataset Specification

6.1 Source Overview

Attribute	Detail
Name	UCI Online Retail II
Repository	UCI Machine Learning Repository (ID: 502)
Origin	Real UK-based non-store online retailer; all-occasion gift-ware; many wholesale customers
Licence	Creative Commons Attribution 4.0 International (CC BY 4.0)
Period	01 December 2009 – 09 December 2011
Total records	~1,067,371 transaction lines (two Excel sheets)
Unique customers	~4,372 (with valid CustomerID)
Unique products	~4,070 stock codes
Countries	38 (UK = 91.4% of transactions)
Citation	Chen, D., Sain, S.L., Guo, K. (2012). Journal of Database Marketing & Customer Strategy Mgmt, 19(3), 197–208

6.2 Data Schema

Column	Type	Description	Known Quality Issues
Invoice	Nominal	6-digit transaction ID; prefix ‘C’ = cancellation	Must flag and separate cancellations
StockCode	Nominal	5-digit product code	Contains non-product operational codes
Description	Text	Product name	~0.27% missing; operational text entries
Quantity	Integer	Units per line item	Negative = returns/cancellations
InvoiceDate	Datetime	Transaction timestamp	Clean
Price	Float	Unit price in GBP (£)	Some zero/negative; inconsistencies per StockCode
Customer ID	Integer	Unique customer identifier	~25% null — excluded from analysis
Country	Text	Customer residence	Clean; 38 unique values

6.3 Selection Rationale

Criterion	Assessment	Status
Real company data (not synthetic)	Genuine UK retailer transactions	✅
Supports full ML spectrum	Clustering, classification, regression, ensembles	✅
Temporal depth for train/holdout split	2-year window enables realistic temporal validation	✅
Business-interpretable features	All columns carry clear business meaning	✅
Academic precedent	Chen et al. (2012) — original RFM segmentation study	✅

Sufficient scale	1M+ transactions, 4K+ customers	✅
Real-world quality challenges	Missing values, returns, dirty entries — demonstrates data engineering	✅

7. Approach & Methodology

7.1 Analytical Framework

The project follows a five-phase analytical chain, where each phase produces outputs consumed by the next. This mirrors the enterprise analytics lifecycle:

Phase	Name	Purpose	Analytics Paradigm
1	Foundation	Ingest, clean, explore, engineer features	Descriptive (Visibility)
2	Intelligence	Segment customers; define churn label	Customer Intelligence (Unsupervised)
3	Prediction	Train and validate churn + CLV models	Predictive (Supervised + Tree-based)
4	Insight	Diagnose drivers; quantify business impact	Diagnostic + Prescriptive-lite
5	Delivery	Build dashboard; document workflow; package portfolio	Communication + Operationalisation

7.2 Modelling Strategy

Task	Approach	Algorithms	Primary Evaluation Metric
Segmentation	Unsupervised clustering on scaled features	K-Means (primary), DBSCAN, Agglomerative	Silhouette score + business interpretability
Churn Prediction	Binary classification with temporal holdout	LogReg (baseline), Random Forest, XGBoost	AUC-ROC + precision-recall at optimised threshold
CLV Prediction	Regression with temporal holdout	Ridge (baseline), RF Regressor, XGBoost Regressor	R^2, MAE; fallback: classification if $R^2 < 0.50$
Interpretation	Post-hoc model explanation	SHAP (global + local + per-segment)	Business translatability of top drivers

7.3 Validation Strategy

Four principles govern all modelling work:

- **Temporal split:** Training on Dec 2009 – Jun 2011; holdout on Jul – Dec 2011. No future information leaks into training features.
- **Stratified cross-validation:** 5-fold StratifiedKFold within training set for hyperparameter tuning; preserves class distribution.
- **Business-relevant thresholds:** Classification threshold optimised for maximum expected profit (not maximum accuracy).
- **Baseline comparison:** Every model benchmarked against a naive baseline. No model accepted unless it meaningfully outperforms the baseline.

7.4 Research-Before-Code Protocol

A strict rule governs this project: no code is written for any deliverable until the corresponding research item is completed. Ten research items (R1–R10) are defined in the Resource Requirements section. Each must produce documented findings before implementation begins.

This protocol mirrors the consulting standard where engagement teams conduct discovery and analysis before solution design.

8. Work Breakdown Structure

All effort estimates assume research is completed before coding begins. Estimates include development, testing, and documentation.

8.1 Phase 1: Foundation (Data Engineering & Exploration)

ID	Task	Effort (h)	Dependency	Output
1.1	Dataset acquisition and loading	1	—	Raw merged dataframe
1.2	Data quality assessment and profiling	3	1.1	Quality report (nulls, types, distributions)
1.3	Data cleaning pipeline	5	1.2	Clean dataframe + reusable functions
1.4	Exploratory data analysis	6	1.3	EDA notebook with key findings
1.5	Feature engineering: RFM	4	1.3	Customer-level RFM table
1.6	Feature engineering: extended behavioural	6	1.3	Extended feature table
1.7	Feature distribution analysis and transformation	2	1.5, 1.6	Documented transformation decisions
Phase 1 Total		**27**		

8.2 Phase 2: Customer Intelligence

ID	Task	Effort (h)	Dependency	Output
2.1	Feature scaling and preprocessing	2	1.7	Scaled feature matrix
2.2	Optimal cluster selection (elbow + silhouette + DBI)	4	2.1	Optimal K with documented rationale
2.3	K-Means clustering	2	2.2	Cluster assignments
2.4	Alternative clustering (DBSCAN, Agglomerative)	3	2.1	Comparison table; selection rationale
2.5	Segment profiling and business labelling	4	2.3	Segment profile cards with names
2.6	Statistical validation across segments	3	2.5	Test results (p-values, effect sizes)
2.7	Churn definition research and threshold testing	3	1.4	Churn definition + sensitivity analysis
2.8	Churn label engineering	2	2.7, 1.6	Labelled customer dataset
Phase 2 Total		**23**		

8.3 Phase 3: Prediction

ID	Task	Effort (h)	Dependency	Output
3.1	Temporal train/test split with leakage verification	2	2.8	Validated train and holdout sets
3.2	Class imbalance strategy selection	3	3.1	Selected technique with rationale
3.3	Churn: Logistic Regression (baseline)	3	3.2	Baseline model + odds ratio table
3.4	Churn: Random Forest (tuned)	4	3.2	Tuned model + metrics
3.5	Churn: XGBoost (tuned)	4	3.2	Tuned model + metrics
3.6	Churn model comparison and selection	3	3.3–3.5	Comparison table; production model selected
3.7	Threshold optimisation for maximum profit	3	3.6	Optimal threshold + profit curve
3.8	CLV: Ridge Regression (baseline)	3	3.1	Baseline CLV model
3.9	CLV: Random Forest Regressor (tuned)	3	3.1	Tuned regressor
3.10	CLV: XGBoost Regressor (tuned)	3	3.1	Tuned regressor
3.11	CLV model comparison and selection	2	3.8–3.10	Selected CLV approach (regression or classification fallback)
Phase 3 Total		**33**		

8.4 Phase 4: Insight & Recommendation

ID	Task	Effort (h)	Dependency	Output
4.1	SHAP: global feature importance	4	3.6	Global SHAP visualisations
4.2	SHAP: per-segment churn driver analysis	4	4.1, 2.5	Top 5 drivers per segment
4.3	SHAP: local customer explanations	2	4.1	Force plots for example customers
4.4	Retention priority matrix construction	3	3.6, 3.11	4-tier matrix with customer counts
4.5	Financial impact and ROI quantification	4	4.4	ROI estimate with stated assumptions
4.6	Segment action recommendation cards	3	4.2, 4.5	Intervention recommendations per segment
Phase 4 Total		**20**		

8.5 Phase 5: Delivery

ID	Task	Effort (h)	Dependency	Output

5.1	Dashboard data preparation and export	2	Phase 4	Dashboard-ready CSV files
5.2	Executive dashboard build (Tableau / Power BI)	8	5.1	Published interactive dashboard
5.3	Workflow architecture diagram	2	All	Pipeline diagram (PNG/SVG)
5.4	Code refactoring into reusable modules	4	All	Clean src/ directory
5.5	README and project narrative	3	All	GitHub README.md
5.6	Results summary (interview leave-behind)	2	All	1-page results PDF
5.7	Final quality review and sign-off	3	5.2–5.6	All quality gates passed
Phase 5 Total		**24**		

8.6 Effort Summary

Phase	Description	Effort (hours)	Cumulative
1	Foundation (Data Engineering & Exploration)	27	27
2	Customer Intelligence (Segmentation)	23	50
3	Prediction (Churn & CLV Modelling)	33	83
4	Insight & Recommendation	20	103
5	Delivery (Dashboard & Portfolio)	24	127
TOTAL		**127**	

9. Milestone Schedule & Quality Gates

9.1 Milestone Schedule

Milestone	Phase	Deliverables	Gate
M1: Data Ready	End Phase 1	D1, D2, D3 — clean data, EDA, feature tables	QG1
M2: Segments Defined	End Phase 2	D4, D5, D6 — validated segments, churn labels	QG2
M3: Models Validated	End Phase 3	D7, D8 — selected churn + CLV models on holdout	QG3
M4: Insights Complete	End Phase 4	D9, D10, D11 — SHAP, priority matrix, ROI	QG4
M5: Project Delivered	End Phase 5	D12, D13 — dashboard, documentation, portfolio	QG5

9.2 Quality Gate Criteria

Each gate is binary: Pass or Fail. All criteria must pass before proceeding to the next phase.

QG1 — Data Ready

#	Check	Pass Criteria
1	No nulls in analysis-critical columns after cleaning	0 nulls in CustomerID, InvoiceDate, Price, Quantity for retained records
2	Non-product rows removed	Zero rows with non-product StockCodes
3	Feature table is customer-level	Row count = unique CustomerID count
4	No temporal data leakage	All features computed from data before Jun 2011 cutoff only

QG2 — Segments Defined

#	Check	Pass Criteria
1	Segments statistically distinct	Kruskal-Wallis $p < 0.05$ on ≥3 RFM dimensions
2	Segments business-interpretable	Each segment named and profiled in plain English
3	Churn label distribution documented	Class ratio reported; imbalance ratio quantified
4	Churn definition sensitivity tested	Results compared across ≥2 inactivity thresholds

QG3 — Models Validated

#	Check	Pass Criteria
1	Churn model AUC-ROC meets target	≥ 0.80 on temporal holdout
2	Churn model outperforms naive baseline	AUC improvement ≥ 0.15 over majority-class
3	CLV model meets target	$R^2 \geq 0.50$ or classification accuracy ≥ 0.75 if pivoted
4	No data leakage detected	Training features verified independent of holdout
5	Model comparison documented	All models compared on all metrics in a single table

QG4 — Insights Complete

#	Check	Pass Criteria
1	Churn drivers identified per segment	≥5 drivers per segment with SHAP + business explanation
2	Priority matrix populated	4 tiers (Protect/Nurture/Monitor/Accept) with counts + £ value
3	ROI is positive	Estimated retention value > estimated campaign cost
4	All ROI assumptions documented	Retention rate, campaign cost, discount rate stated

QG5 — Project Delivered

#	Check	Pass Criteria
1	Dashboard loads and renders correctly	All visualisations display; data matches notebooks
2	5-second rule passes	Key insight visible within 5 seconds (non-technical reviewer)
3	Code is reproducible	git clone + pip install + notebook run = same results
4	README complete	Business context, methodology, findings, reproduction instructions
5	Workflow diagram complete	Full pipeline from data to dashboard with feedback loop

10. Governance Structure

10.1 Roles & Responsibilities

Role	Assigned To	Responsibility
Project Owner	Mai	All decisions, execution, quality assurance, final sign-off
Technical Advisor	Claude (AI)	Research support, methodology guidance, code review, pattern validation
Domain Reviewer	TBD (peer / faculty)	Dashboard usability test, business narrative credibility review

10.2 RACI Matrix

R = Responsible | A = Accountable | C = Consulted | I = Informed

Deliverable	Mai	Claude	Reviewer
D1–D3: Data pipeline + features	R, A	C	I
D4–D6: Segmentation + churn labels	R, A	C	I
D7–D8: Prediction models	R, A	C	I
D9–D11: SHAP + priority + ROI	R, A	C	I
D12: Executive dashboard	R, A	I	C
D13: Documentation + portfolio	R, A	C	I
Quality gate reviews	R, A	C	I

10.3 Decision Authority

Decision Type	Authority	Process
Scope changes	Mai	Change control process (Section 15)
Methodology pivots (e.g., CLV regression → classification)	Mai (after consulting Claude)	Documented rationale + charter update
Timeline adjustments	Mai	Milestone schedule updated
Tool or library substitutions	Mai	Resource plan updated

11. Resource Requirements

11.1 Technical Stack

Layer	Resources	Purpose in Project
Environment	Python 3.10+, Jupyter Lab, Git/GitHub	Development, version control, portfolio hosting
Data Processing	pandas ≥2.0, numpy ≥1.24, datetime	Transaction aggregation, RFM, temporal joins
EDA & Statistics	matplotlib ≥3.7, seaborn ≥0.12, scipy.stats ≥1.10	Distributions, hypothesis testing, visualisation
Unsupervised ML	sklearn.cluster, sklearn.preprocessing, yellowbrick ≥1.5	K-Means, DBSCAN, Agglomerative; scaling; elbow/silhouette
Supervised Classification	sklearn (linear_model, svm, neighbors, ensemble), xgboost ≥2.0	LogReg, SVM, KNN, Random Forest, XGBoost for churn
Supervised Regression	sklearn (Ridge, Lasso, ensemble), xgboost ≥2.0	Ridge, RF Regressor, XGBoost Regressor for CLV
Class Imbalance	imbalanced-learn ≥0.11	SMOTE, RandomUnderSampler
Model Evaluation	sklearn.model_selection, sklearn.metrics	StratifiedKFold, GridSearchCV, AUC-ROC, R^2, MAE
Interpretability	shap ≥0.42	Global/local/per-segment SHAP; force plots, beeswarm
BI Output	plotly ≥5.15, Tableau Public or Power BI (free)	Interactive charts, executive dashboard
Documentation	Markdown, draw.io or Mermaid	README, workflow architecture diagrams

11.2 Knowledge Research Agenda

Each research item must be completed before its dependent deliverables. The research-before-code protocol is enforced at every quality gate.

ID	Topic	Key Questions	Feeds
R1	E-commerce data cleaning	Handling missing CustomerIDs? Returns/cancellations? Non-product codes?	D1
R2	RFM framework	Scoring methodology? Enterprise implementations? Extensions beyond basic RFM?	D3
R3	Clustering for skewed data	K-Means vs. DBSCAN vs. hierarchical? Log transforms? Validation beyond silhouette?	D4
R4	Non-contractual churn	Defining churn without subscriptions? Inactivity threshold? Sensitivity to definition?	D6
R5	Class imbalance	SMOTE vs. undersampling vs. cost-sensitive? Business-optimal threshold selection?	D7
R6	CLV modelling	Historical vs. predictive CLV? BG/NBD awareness? Feature drivers?	D8

R7	SHAP for business	Translating SHAP to plain English? Global vs. local? Stakeholder presentation?	D9
R8	Dashboard design	Big 4 dashboard standards? 5-second rule? Customer analytics KPIs?	D12
R9	Pipeline architecture	Enterprise pipeline structure? Documentation standards? Retraining cadence?	D13
R10	Retention economics	Cost of churn formulas? Campaign ROI methodology? Consulting ROI presentation?	D11

12. Risk Register

12.1 Risk Scoring Matrix

Likelihood (L) and Impact (I) are each scored 1–5. Risk Score = L × I. Scores ≥12 require active mitigation; 6–11 require monitoring; ≤5 are accepted.

	Impact 1 (Negligible)	Impact 2 (Minor)	Impact 3 (Moderate)	Impact 4 (Significant)	Impact 5 (Severe)
L5: Almost Certain	5	10	15	20	25
L4: Likely	4	8	12	16	20
L3: Possible	3	6	9	12	15
L2: Unlikely	2	4	6	8	10
L1: Rare	1	2	3	4	5

12.2 Risk Log

ID	Risk	L	I	Score	Mitigation	Owner
RK-01	Churn definition ambiguity in non-contractual setting distorts labels and model performance	4	4	16	Test 30/60/90-day thresholds; report sensitivity; cite academic justification	Mai
RK-02	Class imbalance biases models toward majority class	4	3	12	SMOTE + cost-sensitive learning + threshold optimisation; precision-recall reporting	Mai
RK-03	Project indistinguishable from standard Kaggle notebooks	4	5	20	Differentiate via workflow structure, per-segment SHAP, ROI quantification, BI dashboard, production-grade docs	Mai
RK-04	CLV regression yields $R^2 < 0.50$	3	3	9	Pre-planned pivot to high/medium/low classification; documented decision	Mai
RK-05	Data leakage from holdout into training	2	5	10	Strict temporal cutoff; verification at QG1 and QG3	Mai
RK-06	Scope creep beyond charter boundaries	3	3	9	Change control process; modular phased design	Mai
RK-07	Time overrun (127h alongside 4 courses + TA)	3	4	12	Phased delivery; Phase 5 can be simplified if needed	Mai
RK-08	25% null CustomerID introduces survivorship bias	3	2	6	Compare null vs. known behaviour in EDA; document limitations	Mai

13. Assumptions & Dependencies

13.1 Assumptions

ID	Assumption	Risk if Invalid	Validation
A1	Customers with zero purchases in holdout period are churned	Mislabelling distorts training	Sensitivity across multiple thresholds (RK-01)
A2	UK-only filtering does not introduce significant bias	Findings may not generalise	Compare UK vs. non-UK in EDA
A3	RFM + extended features provide sufficient signal	Models underperform	Monitor QG3; extend features if needed
A4	Null CustomerID records are missing at random	Survivorship bias	Statistical comparison in EDA
A5	Campaign cost of £10–£50/customer is reasonable for ROI	ROI estimate loses credibility	Benchmark vs. published retail costs; label as assumption
A6	Retention success rate of 10–30% for targeted campaigns	ROI over/underestimated	Cite benchmarks; present ROI as range

13.2 Dependencies

ID	Dependency	Type	Impact if Unmet	Mitigation
DEP-01	UCI dataset remains publicly available	External	No alternative pre-selected	Download and store locally at project start
DEP-02	Python library compatibility	Technical	Low — all stable	Pin versions in requirements.txt
DEP-03	Tableau Public / Power BI free tier	External	Low — both free	Plotly dashboards as fallback
DEP-04	R1–R10 research before implementation	Internal	Phase delays	Research-first rule in quality gates

14. Success Criteria

Success is assessed across three tiers. Tier 1 is the minimum acceptable standard. Tier 3 represents stretch goals that further differentiate the project.

14.1 Tier 1: Minimum Viable (all must be achieved)

ID	Criterion	Measure
SC-1	Customer segments are statistically valid and business-interpretable	≥3 segments, $p < 0.05$ inter-group, plain-English labels
SC-2	Churn model outperforms baseline on holdout	AUC-ROC ≥ 0.80
SC-3	Financial impact is quantified with stated assumptions	ROI ratio expressed
SC-4	Workflow is documented as a repeatable pipeline	Architecture diagram complete

14.2 Tier 2: Target (should achieve most)

ID	Criterion	Measure
SC-5	CLV model explains meaningful variance	$R^2 \geq 0.50$ or classification accuracy ≥ 0.75
SC-6	Per-segment churn drivers identified and explained	≥5 drivers per segment via SHAP in business language
SC-7	Executive dashboard passes 5-second test	Confirmed by non-technical reviewer
SC-8	Code is fully reproducible from a fresh clone	git clone → install → run = same results

14.3 Tier 3: Stretch (differentiators)

ID	Criterion	Measure
SC-9	Threshold optimised for maximum expected profit	Profit curve plotted with business justification
SC-10	4-tier retention priority matrix with financial values	Customer counts and £ per tier
SC-11	Project reads as a consulting deliverable, not academic work	Non-technical peer confirms professional framing

15. Change Control Process

Any modification to scope, timeline, methodology, or deliverables after charter approval must follow this process:

Step	Action	Responsible
1. Identify	Document the proposed change: what, why, impact on scope/timeline/quality	Mai
2. Classify	Minor: no impact on milestones or success criteria. Major: impacts milestones, scope, or criteria.	Mai
3. Assess (Major only)	Document rationale; reassess risk register; confirm timeline impact; consult Claude for methodology	Mai + Claude
4. Implement	Update relevant charter sections; log in Document Control version table	Mai
5. Communicate	Inform stakeholders per Communication Plan	Mai

16. Communication Plan

Communication	Audience	Frequency	Channel	Content
Quality gate review	Mai + Claude	At each milestone	Claude.ai project	Gate criteria pass/fail; blockers; next-phase plan
Phase completion summary	Mai (self-reflection)	At each milestone	Project journal (Markdown)	What worked, what changed, lessons learned
Domain review	Domain Reviewer	Phase 5 (dashboard)	In-person or video	Dashboard usability; business narrative credibility
Portfolio update	Public (GitHub)	Phase 5 (final)	GitHub repository	README, notebooks, dashboard link, results summary

17. Stakeholder Analysis

Stakeholders are mapped on a Power/Interest grid to determine engagement strategy.

Stakeholder	Power	Interest	Quadrant	Strategy
Mai (Project Owner)	High	High	Manage Closely	Full ownership of all decisions and execution
Claude (Technical Advisor)	Low	High	Keep Informed	Consulted on methodology; informed of decisions
Big 4 Interviewers (future)	High	Low	Keep Satisfied	Project designed to meet their evaluation criteria
Domain Reviewer	Low	Medium	Monitor	Engaged at Phase 5 for usability review
University Faculty	Medium	Low	Keep Satisfied	Project demonstrates competency beyond coursework

18. Portfolio Output Structure

18.1 Final Deliverable Artefacts

#	Artefact	Format	Primary Audience
1	GitHub Repository	Code + documentation	Technical reviewers (DS hiring managers)
2	Jupyter Notebooks (modular, one per phase)	.ipynb	Technical reviewers inspecting methodology
3	Executive Dashboard	Tableau Public / Power BI	Non-technical stakeholders (interviewers, managers)
4	README.md	Markdown	First impression: narrative, context, results
5	Workflow Architecture Diagram	PNG/SVG	Demonstrates production thinking
6	Results Summary	1-page PDF	Interview leave-behind

18.2 Repository Structure

The repository follows a clean, professional layout separating notebooks (analytical narrative), source code (reusable functions), data outputs, and documentation:

Directory	Contents
notebooks/	01_data_ingestion.ipynb through 09_dashboard_export.ipynb
src/	data_cleaning.py, feature_engineering.py, model_training.py, evaluation.py
outputs/	segment_profiles.csv, customer_scores.csv, model_comparison.csv, dashboard_data.csv
figures/	Key visualisations for README and results summary
docs/	Project charter, workflow diagram, results summary PDF
Root	README.md, requirements.txt, .gitignore, LICENSE

19. Glossary

Term	Definition
AUC-ROC	Area Under the Receiver Operating Characteristic Curve — classification performance metric (0.5 = random, 1.0 = perfect)
Churn	A customer ceasing to purchase within a defined time window
CLV	Customer Lifetime Value — predicted total revenue a customer will generate
DBSCAN	Density-Based Spatial Clustering of Applications with Noise — identifies clusters of varying shapes
K-Means	Centroid-based unsupervised clustering that partitions data into K groups
Precision	Of customers predicted as churners, the proportion that actually churned
Quality Gate	Binary checkpoint between phases; all criteria must pass before proceeding
RACI	Responsible, Accountable, Consulted, Informed — role assignment matrix
Recall	Of customers who actually churned, the proportion the model identified
RFM	Recency, Frequency, Monetary — customer segmentation framework based on purchase behaviour
SHAP	SHapley Additive exPlanations — game-theoretic approach to explain individual model predictions
Silhouette Score	Cluster quality metric: similarity to own cluster vs. others (range: −1 to +1)
SMOTE	Synthetic Minority Oversampling Technique — generates synthetic minority-class examples
Temporal Holdout	Validation strategy where test set is chronologically after training, simulating real prediction
WBS	Work Breakdown Structure — hierarchical decomposition of project scope into tasks
XGBoost	Extreme Gradient Boosting — tree-based ensemble; industry standard for tabular prediction

20. Charter Approval

By signing below, the undersigned acknowledge that they have reviewed this charter and approve the project to proceed as defined.

Role	Name	Signature	Date
Project Owner	Mai		
Technical Advisor	Claude		

END OF CHARTER

Upon approval, this charter authorises commencement of Phase 0 (dataset acquisition) and Phase 1 research activities (R1, R2).